Python For Loop

10 beginner-friendly examples with code, explanation, and output

What is a for loop?

A **for loop** is used to repeat a block of code for each item in a sequence, such as a range of numbers, a list, or a string.

Basic syntax

```
for variable in sequence:
    # statements to repeat
```

Important note: In range(start, stop), the start value is included but the stop value is not included.

Example 1: Print numbers from 1 to 5

The range(1, 6) generates numbers from 1 up to 5. The ending value 6 is not included.

Code:

```
for i in range(1, 6):
    print(i)
```

Output:

```
1
2
3
4
5
```

Example 2: Print even numbers from 2 to 10

The loop checks each number from 2 to 10. The if condition prints only numbers divisible by 2.

Code:

```
for i in range(2, 11):
    if i % 2 == 0:
        print(i)
```

Output:

```
2
4
6
8
10
```

Example 3: Print odd numbers from 1 to 9

A number is odd when its remainder after division by 2 is 1.

Code:

```
for i in range(1, 10):
    if i % 2 != 0:
        print(i)
```

Output:

```
1
3
5
```

```
7
9
```

Example 4: Find the sum of numbers from 1 to 5

The variable total starts at 0. Each loop adds the current number to total.

Code:

```
total = 0
for i in range(1, 6):
    total = total + i
print("Sum =", total)
```

Output:

```
Sum = 15
```

Example 5: Print a multiplication table

This example prints the table of 5 from 1 to 10. The f-string displays the calculation clearly.

Code:

```
num = 5
for i in range(1, 11):
    print(f"{num} x {i} = {num * i}")
```

Output:

```
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
...
5 x 10 = 50
```

Example 6: Find the factorial of a number

Factorial means multiplying all positive integers from 1 to the given number. For 5, it is 5 x 4 x 3 x 2 x 1.

Code:

```
num = 5
factorial = 1
for i in range(1, num + 1):
    factorial = factorial * i
print("Factorial =", factorial)
```

Output:

```
Factorial = 120
```

Example 7: Loop through a list

A for loop can visit each item in a list directly. Here, name takes one fruit at a time.

Code:

```
fruits = ["Apple", "Mango", "Banana"]
for fruit in fruits:
    print(fruit)
```

Output:

```
Apple
Mango
Banana
```

Example 8: Count the number of vowels in a word

The loop checks each character. If it is a vowel, the count increases by 1.

Code:

```
word = "education"
count = 0
for ch in word:
    if ch in "aeiou":
        count += 1
print("Vowels =", count)
```

Output:

```
Vowels = 5
```

Example 9: Print a square pattern

The outer loop runs 3 times. The inner loop prints three stars on each row.

Code:

```
for row in range(3):
    for col in range(3):
        print("*", end=" ")
    print()
```

Output:

```
* * *
* * *
* * *
```

Example 10: Reverse a string

The loop uses a negative step to move from the last character to the first character.

Code:

```
word = "PYTHON"
for i in range(len(word) - 1, -1, -1):
    print(word[i], end="")
```

Output:

```
NOHTYP
```

Quick practice

1. Print numbers from 10 to 1 in reverse order.
2. Find the sum of even numbers from 1 to 20.
3. Count how many letters are in your name.
4. Print the multiplication table of a number entered by the user.

Remember: Indentation is important in Python. The statements inside a for loop must be indented.